

Disproof of conjecture 00000000747

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Abstract

Conjecture 00000000747 asserts that for every prime $p \geq 5$ the fixed points of $x \mapsto x^p$ in the p -adic integers $\mathbb{Z}_p$ are only 0 and 1. We show that the fixed points of $x \mapsto x^p$ on $\mathbb{Z}_p$ are exactly the roots of $f(x) = x^p - x$. By Fermat's little theorem every residue modulo p is a root; since $f'(x) = px^{p-1} - 1 \equiv -1 \pmod{p}$ is a unit, Hensel's lemma lifts each of the p residue classes to exactly one root in $\mathbb{Z}_p$. Hence there are exactly p fixed points, namely $\{0\} \cup \mu_{p-1}$ (the Teichmüller roots). For $p \geq 5$ that is at least 5 fixed points, not 2. The same conclusion holds under the alternative reading " $\mathbb{Z}_p = \mathbb{Z}/p\mathbb{Z}$ ", where $x^p = x$ has all p residues as roots in $\mathbb{F}_p$. Conjecture 00000000747 is therefore false.

1 The conjecture

Conjecture 00000000747. For $p \geq 5$, the iterative fixed points of $x \mapsto x^p$ in $\mathbb{Z}_p$ are only 0 and 1 (rigidity of superattracting orbits).

A fixed point of the map $x \mapsto x^p$ on $\mathbb{Z}_p$ is a solution of $x^p = x$; that is what is analysed below. The word "iterative" and the phrase "superattracting orbits" are discussed in Section 7, where we note that no natural reading rescues the claim.

2 Fixed points are the roots of $f(x) = x^p - x$

Definition 1. Let p be prime and let

$$f(x) = x^p - x \in \mathbb{Z}[x].$$

Write $\mathbb{Z}_p$ for the ring of p -adic integers and $\mathbb{F}_p = \mathbb{Z}/p\mathbb{Z}$ for the residue field.

Proposition 2. For $x \in \mathbb{Z}_p$ one has $x^p = x$ if and only if $f(x) = 0$. Consequently the fixed points of $x \mapsto x^p$ in $\mathbb{Z}_p$ are exactly the roots of f in $\mathbb{Z}_p$, and the fixed points modulo p are exactly the roots of $\bar{f}(x) = x^p - x$ in $\mathbb{F}_p$.

Proof. Immediate from the definition of f and reduction modulo p . $\square$

3 Fermat: every residue is a root modulo p

Theorem 3 (Fermat's little theorem). For every $a \in \mathbb{F}_p$ one has $a^p = a$. Equivalently,

$$a^p - a \equiv 0 \pmod{p} \quad \text{for all } a \in \mathbb{Z}.$$

Proof. The multiplicative group $\mathbb{F}_p^\times$ has order $p - 1$, so $a^{p-1} = 1$ for every $a \neq 0$, and multiplying by a gives $a^p = a$; the case $a = 0$ is immediate. The integer form follows by reducing a modulo p . $\square$

Corollary 4. The polynomial $f(x) = x^p - x$ has all p elements of $\mathbb{F}_p$ as roots, and

$$f(x) = x^p - x = \prod_{a \in \mathbb{F}_p} (x - a) \quad \text{in } \mathbb{F}_p[x].$$

In particular, for $p \geq 5$ there are at least 5 roots modulo p — already more than 0 and 1.

Proof. The first statement is Theorem 3. The displayed factorisation follows because both sides are monic of degree p and the left side vanishes at all p roots $a \in \mathbb{F}_p$, whose number equals the degree; equivalently, $x^p - x$ is the product of the distinct linear factors $x - a$. $\square$

4 The derivative and Hensel lifting

Lemma 5 (Derivative computation). For $f(x) = x^p - x$ one has

$$f'(x) = p x^{p-1} - 1 \equiv -1 \pmod{p}.$$

Hence $f'(a) \not\equiv 0 \pmod{p}$ for every $a \in \mathbb{F}_p$: all roots of f modulo p are simple, and $f'(a)$ is a unit in $\mathbb{Z}_p$.

Proof. Differentiate $x^p - x$ termwise. Since $p \equiv 0 \pmod{p}$, the term $p x^{p-1}$ vanishes modulo p , leaving -1 , which is a unit of $\mathbb{F}_p$ and of $\mathbb{Z}_p$. $\square$

Lemma 6 (Hensel, simple-root form). *Let $g \in \mathbb{Z}[x]$ and let $a \in \mathbb{Z}$ satisfy $g(a) \equiv 0 \pmod{p}$ and $g'(a) \not\equiv 0 \pmod{p}$. Then there is a unique $\alpha \in \mathbb{Z}_p$ with*

$$\alpha \equiv a \pmod{p}, \quad g(\alpha) = 0.$$

Equivalently, each residue class $a \in \mathbb{F}_p$ that is a simple root of $\bar{g}$ contains exactly one root of g in $\mathbb{Z}_p$.

Proof. This is the standard simple-root case of Hensel's lemma. Briefly, given a root a_k modulo p^k one seeks $a_{k+1} = a_k + t p^k$ with $t \in \{0, \dots, p-1\}$ and $g(a_{k+1}) \equiv 0 \pmod{p^{k+1}}$. Expanding,

$$g(a_k + t p^k) \equiv g(a_k) + t p^k g'(a_k) \pmod{p^{k+1}},$$

and since $g(a_k) = c p^k$ for some $c \in \mathbb{Z}$ while $g'(a_k) \equiv g'(a) \not\equiv 0 \pmod{p}$ is invertible, the congruence $c + t g'(a_k) \equiv 0 \pmod{p}$ determines t uniquely modulo p . The resulting compatible sequence $(a_k)_{k \geq 1}$ converges in the complete ring $\mathbb{Z}_p$ to the unique root α with $\alpha \equiv a \pmod{p}$ (and $\alpha \equiv a_k \pmod{p^k}$ for every k). $\square$

Theorem 7 (Main count). *Let p be prime. The fixed points of $x \mapsto x^p$ in $\mathbb{Z}_p$ are exactly*

$$\{x \in \mathbb{Z}_p : x^p = x\} = \{0\} \cup \mu_{p-1},$$

where $\mu_{p-1} = \{\zeta \in \mathbb{Z}_p : \zeta^{p-1} = 1\}$ is the group of $(p-1)$ -st roots of unity (the Teichmüller roots). There are exactly p of them: one in each residue class modulo p .

Proof. By Proposition 2 the fixed points are the roots of $f(x) = x^p - x = x(x^{p-1} - 1)$, so $x = 0$ or $x^{p-1} = 1$; this gives the description $\{0\} \cup \mu_{p-1}$. Corollary 4 and Lemma 5 show that every one of the p residue classes modulo p is a simple root of f modulo p . Applying Lemma 6 to each class produces exactly one root in $\mathbb{Z}_p$ per class, and these are all the roots because any root reduces to some class. Hence the root set has exactly p elements, namely 0 together with the $p-1$ elements of μ_{p-1} (equivalently, the Teichmüller lifts of $\mathbb{F}_p^\times$). $\square$

Corollary 8. *For every prime $p \geq 5$ the fixed point set of $x \mapsto x^p$ in $\mathbb{Z}_p$ has exactly $p \geq 5$ elements. In particular it is not contained in $\{0, 1\}$, and conjecture 00000000747 is false.*

Proof. By Theorem 7 the fixed point set has p elements. For $p \geq 5$ we have $p \geq 5 > 2$, while $\{0, 1\}$ has two elements; the nonzero elements of μ_{p-1} other than 1 are additional fixed points. Hence the fixed points are not only 0 and 1. $\square$

5 The explicit example $p = 5$

Take $p = 5$, so $f(x) = x^5 - x$ and $f'(x) = 5x^4 - 1 \equiv -1 \pmod{5}$. Every residue class modulo 5 is a simple root of f modulo 5, and each lifts uniquely to a root in $\mathbb{Z}_5$.

The lift of the class 2 mod 5. Iterating the Hensel step from $a_1 = 2$ gives the compatible sequence

$$2, 7, 57, 182, 2057, 14557, 45807, 280182 \pmod{5, 5^2, \dots, 5^8},$$

i.e. the unique root of f in $\mathbb{Z}_5$ with $x \equiv 2 \pmod{5}$ is

$$\alpha = 280182 \pmod{5^8},$$

and indeed

$$280182^5 \equiv 280182 \pmod{5^8 = 390625}.$$

This is neither 0 nor 1: it is $\equiv 2 \pmod{5}$, whereas $0 \equiv 0$ and $1 \equiv 1 \pmod{5}$, and a congruence modulo 5^8 would imply the same congruence modulo 5.

A second explicit non-trivial root. The integer 110443 satisfies

$$110443^5 \equiv 110443 \pmod{5^8}, \quad 110443 \not\equiv 0, 1 \pmod{5^8},$$

so $110443 \pmod{5^8}$ is a further non-trivial root. Its residue modulo 5 is $110443 \equiv 3 \pmod{5}$; thus 110443 is the Hensel lift of the class 3 mod 5 (the lift of class 2 being 280182 above). Either value already contradicts “only 0 and 1”.

In summary, modulo 5^8 the roots of $x^5 = x$ include 0, 1, 280182 and 110443, and there are exactly 5 roots in all, one in each class modulo 5:

$$\{0\} \cup \mu_4, \quad |\mu_4| = 4.$$

There is no way to single out $\{0, 1\}$.

6 The alternative reading $\mathbb{Z}_p = \mathbb{Z}/p\mathbb{Z}$

If instead “ $\mathbb{Z}_p$ ” is read as the finite ring $\mathbb{Z}/p\mathbb{Z} = \mathbb{F}_p$, then the fixed points of $x \mapsto x^p$ in $\mathbb{F}_p$ are the roots of $x^p - x$, which by Theorem 3 are *all* p residues. For $p \geq 5$ there are at least 5 fixed points, again not only 0 and 1. Thus the conjecture fails under its finite reading as well.

7 On the iterative and superattracting language

The conjecture speaks of “iterative fixed points” and of “rigidity of superattracting orbits”. We record that no natural reading rescues the claim.

Remark 9 (Iterates). If “iterative fixed points of $x \mapsto x^p$ ” is read as the periodic points fixed by the n -th iterate $x \mapsto x^{p^n}$, the fixed point equation becomes $x^{p^n} = x$, whose roots in $\mathbb{Z}_p$ are $x = 0$ together with the roots of $x^{p^n-1} = 1$. Since $\gcd(p, p^n - 1) = 1$, the group of roots of unity of order dividing $p^n - 1$ sits inside $\mathbb{Z}_p^\times$ and has $p^n - 1$ elements, so there are p^n such points, again one per residue class modulo p . This set is even larger than the fixed point set, so the claim fails a fortiori.

Remark 10 (Superattracting). The map under iteration is $g(x) = x^p$, whose multiplier at a fixed point x is

$$g'(x) = p x^{p-1}.$$

At $x = 0$ this is 0, so 0 is indeed superattracting. At any $x \in \mu_{p-1}$ one has $x^{p-1} = 1$, hence $g'(x) = p$ with $|p|_p = p^{-1} < 1$: such a point is *attracting but not superattracting*. Thus “superattracting” selects only 0 among the fixed points and does not single out the pair $\{0, 1\}$; in particular it cannot justify the claim that 1 is the only other fixed point (or any other restriction). The dynamical vocabulary does not change the algebraic count of p fixed points.

8 Formal verification

The full statement that the fixed point set has exactly p elements relies on Hensel’s lemma. Core Lean 4 does not provide the p -adic integers, so the Lean development in `lean4/` formalises the *finite residue shadow* of the argument:

- `fermat_five` and `roots_mod_5`: all five residues modulo 5 satisfy $x^5 = x$ (Theorem 3);
- `nontrivial_root_mod_5_pow_8`: the explicit root $110443^5 \equiv 110443 \pmod{5^8}$, neither 0 nor 1;
- `hensel_step_k1`: one concrete Hensel lifting step ($k = 1 \rightarrow k = 2$ at $p = 5$), the finite instance of Lemma 6;
- `conjecture_00000000747_false`: the packaged finite disproof.

The general Hensel step and its consequence for $\mathbb{Z}_p$ are proved above and are additionally verified numerically by `reproduce.py`, which counts exactly $p = 5, 7, 11$ roots modulo p^k for all $k \leq 8$. See `lean4/README.md` for the precise scope.

9 Conclusion

The fixed points of $x \mapsto x^p$ on $\mathbb{Z}_p$ are the roots of $x^p - x$. Modulo p all p residue classes are simple roots by Fermat's little theorem, and Hensel's lemma lifts each class to exactly one root in $\mathbb{Z}_p$. Hence the fixed point set is $\{0\} \cup \mu_{p-1}$ and has exactly p elements. For $p \geq 5$ this is at least 5, not the two points 0 and 1 claimed. The same count refutes the modulo- p reading. Conjecture 00000000747 is therefore **false**.